CART-based optimization of core and cladding layers in silicon nitride photonic integrated circuits towards propagation and bend excess loss minimization

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ABSTRACT

In the burgeoning field of sensing, Photonic Integrated Circuits (PICs) are essential tools for precise, high-speed detection of biological markers and particles. The performance of these biosensors is intricately linked to the losses of PICs, which is largely determined by the configuration of their core and cladding layers. Recognizing this, the present study ventures into the optimization of these layers in Silicon Nitride (Si_3N_4) PICs, employing an innovative approach using Classification and Regression Trees (CART). The study identifies propagation and bend excess losses, two critical factors affecting PIC performance, as response variables. In contrast, the physical characteristics of the core and cladding layers are considered as input variables. To ensure the robustness and completeness of the study, an appropriate Design of Experiments (DOE) was implemented, meticulously exploring possible combinations of layer configurations. Following the DOE, the CART algorithm was then applied to this design space, where the losses act as response variables. The algorithm functions by partitioning the design space into regions associated with specific layer configurations and iteratively refines these partitions based on their corresponding impact on propagation and bend excess losses. The end results of this process is statistical information about the layer stacks which come with significantly low propagation and bend excess losses, thereby enhancing PIC performance. This improvement in performance directly translates to heightened sensitivity and specificity in biosensors. Furthermore, the successfull application of the CART methodology has demonstrated its potential to streamline the PIC design process, enhancing its robustness, an aspect critical for practical implementation in fabrication environments.

Keywords: Silicon nitride waveguides, Classification and regression trees, Design of experiment, Loss optimization, NIR, Sensing

1. INTRODUCTION

The realm of integrated photonics, which involves manipulating and propagating light on micro and nano scales, has experienced a transformative ascent in recent years. From high-speed telecommunications and advanced computing to intricate sensing applications, integrated photonics has laid the foundation for many innovations.[1,2] At the core of this technological evolution are Photonic Integrated Circuits (PICs), which have come to redefine the dynamics of high-speed detection in biosensing and beyond (Smith et al., 2017). Despite such milestones, these systems are subject to a persistent challenge: robust optical loss minimization in devices for mass fabrication, primarily originating from surface-roughness-induced scattering and absorption losses.[3,4]

The study involves Si_3N_4 waveguides, which are recognized for their compatibility with low-temperature back-end and high-temperature front-end processes. The modelling approach was evaluated across different waveguide widths and bend radii in the infrared region. The aim of using rigorous examination, combined with an automated wafer-level measurement setup, was to provide a granular understanding of the interplay among different layer

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combinations of the Si_3N_4 and cladding layers which can be used for different processes optimized applications, like sensing or spectroscopy.[5,6]

In the case studied in this work, the design of Experiments (DOE) is paramount in investigating the potential effects and interactions of multiple variables. For wafer-level testing of the integrated structures, especially when examining propagation and bend excess losses, the full-factorial design of the involved layers proved to be particularly insightful. Every possible combination of factors and their levels were tested in this experimental setup. Such a meticulous approach enables researchers to discern the primary effects of individual factors and their interaction effects on propagation and bend excess losses. Consequently, this methodology facilitates the identification of optimal configurations and the conditions under which PICs achieve minimal losses. The wafer-level testing, in combination with the full-factorial DOE, ensures that results are not only statistically robust but also practically significant for mass fabrication applications in the vast and complex domain of integrated photonics.[7,8]

Navigating the intricacies of PIC design demands a profound understanding of photonics and the leveraging of robust statistical methodologies. In addressing this multifaceted challenge, this study uses an integrative approach, harmonizing the foundations of traditional photonics research with the prowess of statistical learning epitomized by classification and regression trees (CART).

With its non-linear, decision tree-based algorithm, CART emerges as a formidable tool in this scenario. At its core, CART is equipped to discern pivotal thresholds within predictor variables deftly. This inherent capability renders it uniquely positioned to delineate the complex relationships in PIC design, accentuating, in particular, the nuanced influence of the layer stack on the losses. Notably, when exploring the layers of a PIC - the undercladding, the central nitride core and the uppercladding - CART's methodology can segregate and classify these layers, treating them as predictor variables. The propagation and bend excess losses, inherently stemming from these layered configurations, depending on layer properties like the roughness of the layer, its absorption, thickness and the real part of the refractive index, were used as response variables. However, roughness, absorption and refractive index were not used as predictor variables directly. The regression analysis offers insights into how each layer's specific attributes, both singularly and in interaction with one another, contribute to the overall propagation and bend excess losses of the PIC.[9,10]

In essence, the aim of the work for this paper was not just to elucidate the optical loss dependence of the layer but to pioneer an innovative methodology, offering a nuanced comparison of measured losses against predicted ones. Through the combined usage of CART and DOE, we aim to furnish a holistic understanding of the layers' roles and their intricate interplays for advances in waveguide design mass fabrication processes.

2. DESIGN AND LAYOUT OF TEST STRUCTURES

To properly use the losses as response variables, one has to design appropriate test structures. The target wavelength for later usage for the platform in the sensing application is around 850 nm and the target polarization is transverse-electric (TE). Waveguide losses stem from various factors, including material absorption and scattering between the waveguide core and its cladding via the roughnesses of the waveguide layers. For standard passive entities like strip waveguides, interface scattering, primarily driven by surface roughness, reigns supreme, especially for lower widths at the targeted waveguide height of 250 nm.[1]

In anticipation of leveraging the design for later-stage chemo/biosensing applications, waveguide widths ranging from 350 nm to 750 nm were investigated. When combined with the specified height and operating wavelength, these widths function within the single-mode regime and are recognized for their minimal propagation losses. Moreover, they exhibit an enhanced field maximum surrounding the waveguide core during TE polarization, which augments the reliability and precision of biosensing measurements.[11]

Custom test structures were designed utilizing Synopsis Optodesigner. The propagation loss was determined using the cutback method with waveguide spirals of 4 different lengths and a constant bend radii of 50 µm. An example of such a cutback structure can be seen in Figure 1.

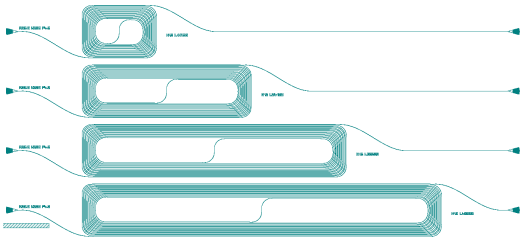


Figure 1: Test structures used to assess the propagation loss corresponding to structures with a successive length difference of 1.24 cm and widths of 350, 450, 550, 590, 650 and 750 nm. The radius of the spirals is 50 µm. Each of these test structures was equipped with grating couplers to assist in interfacing a power meter and a diode laser that emitted TE-polarized light at 850 nm (Depicted example array for waveguide width of 350 nm.)

The bend excess loss was evaluated using spirals with radii of 10, 12.5, 15, 17.5, 20, 25, 30 and 35 µm. These were contrasted with a structure with a radius of 70 µm, which is expected to demonstrate comparably low bend excess losses regarding the other radii. All the structures in the figure have a consistent path length of approximately 0.92 cm and widths of 450, 550, 590, 650 and 750 nm. Visual representations of these bend excess loss structures can be found in Figure 2.

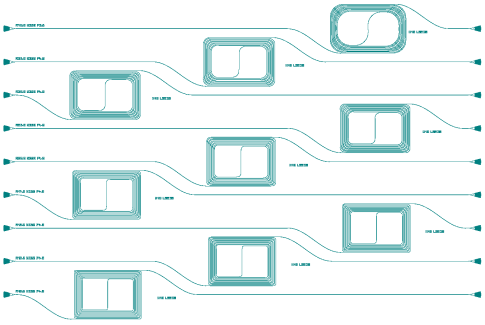


Figure 2: Test structures used to assess the bend excess loss corresponding to structures with radii of 10, 12.5, 15, 17.5, 20, 25, 30 and 35 µmand widths of 450, 550, 590, 650 and 750 nm. The radius of the reference spiral is 70 µm. Each of these test structures was equipped with grating couplers to assist in interfacing a power meter and a diode laser that emitted TE-polarized light at 850 nm (Depicted example array for waveguide width of 350 nm).

3. DESIGN OF EXPERIMENT FOR FABRICATION

The general layer stack comprised of structures made from silicon nitride (Si_3N_4) waveguides set within a silicon dioxide (SiO_2) undercladding and an uppercladding of the same material. The deposition steps were conducted for the tree-based modelling via different techniques in a full-factorial DOE, a systematic method used to understand the influence of multiple factors on a specified outcome by examining all possible combinations. For layer stacks, this would involve studying the effect of each layer's material and the process conditions under which they are formed on desired outcomes, such as propagation losses and bend excess losses. In this design, the three layers (factors) were combined with two variations in the deposition of each layer (level), where all combinations were fabricated, each on five wafers, which ensures the minimum statistical sample size, so 40 wafers were investigated. According to the study of Melike Kaya Bahçecitapar et. al., the optimal number of replicates for the 2x3 full-factorial designs, taking into consideration standard deviation and the main difference of the response, would involve more than 10 samples since the maximum difference of the losses is not expected to be that high (order of 1), however, due to cost reasons five were used here.[12] In general, the power of a test, which is the likelihood of identifying real differences, increases for larger values of maximum differences.

The different layers used in the stacks are displayed in Table 1, including the thickness, refractive index (RI) and absorption (Abs) measured using a M-2000UI Woollam ellipsometer and utilizing a Cauchy model, including Urbach absorption. The roughness of the undercladding surface, the waveguide top-surface and the sidewall (measured with a tilting technique) were determined utilizing the Anton Paar Tosca 400 Atomic Force Microscope.[13] As undercladding, either high-density plasma-enhanced chemical vapor deposition (HD-PECVD) SiO_2, deposited at 450 °C substrate temperature, was utilized or thermal SiO_2, grown in a wet chemical process at a 1200 °C substrate temperature in a furnace. The Si_3N_4 waveguide layer was deposited via plasma-enhanced chemical vapour deposition (PECVD) at 400 °C substrate temperature and low-pressure chemical vapour deposition (LPCVD) at 770 °C substrate temperature in a furnace. The final uppercladding layer was then realized as a HD-PECVD SiO_2 under the same conditions as previously mentioned or as sputtered SiO_2 at a substrate temperature of 100 °C using polycrystalline Si as the target material.

Table 1: Thin film layer selection utilized in DOE of the Si_3N_4 waveguide platform. Thickness, refractive index and absorption were determined by the M-2000UI Woollam ellipsometer.

	Thickness	RI @ 850 nm [1]	Abs @ 850 nm [1]
Thermal SiO_2 (Undercladding)	2.2 µm ± 120 nm	1.50 ± 0.05	~ 0
HD-PECVD SiO_2 (Undercladding; after CMP)	2.2 µm ± 220 nm	1.46 ± 0.05	~ 0
HD-PECVD SiO_2 (Uppercladding)	2.4 µm ± 150 nm	1.46 ± 0.05	~ 0
Sputtered SiO_2(Uppercladding)	2.4 µm ± 120 nm	1.48 ± 0.05	~ 0
PECVD Si_3N_4 (Waveguide)	250 nm ± 15 nm	1.91 ± 0.05	$\sim 3 * 10^{-6}$
LPCVD Si_3N_4 (Waveguide)	250 nm ± 15 nm	1.98 ± 0.05	$\sim 1 * 10^{-6}$

The roughness data was then evaluated using the Gwyddion software of the tool and also, for the waveguides by the following steps. The process starts by detrending the data; a plane determined by least squares was subtracted from the data and the sampling grid was adjusted accordingly. The initial model used an anisotropic, separable autocovariance kernel combining exponential and Matérn 3/2 kernel functions. However, due to stability problems with this model, mainly converging to noise at high frequencies, additional elements are included: a 2D exponential kernel and a white noise kernel. The model emphasized the scattering kernel, noting the necessity of an exponential kernel in one dimension for accurate fitting, even though the noise kernel's contribution is minimal. To ensure convergence in the spectral analysis, multitaper functions based on the discrete prolate spheroidal sequence (DPSS) were utilized, extending the 1D DPSS to 2D. This method employed multiple frequency bands for robustness, where each AFM measurement is tapered with 36 tapers and the spectral analysis of each tapered sample was summed using the Whittle likelihood. This approach reflects a thorough strategy to tackle AFM data complexities due to misalignment, focusing on surface and sidewall roughness measurements.[13] The data can be seen in Table 2.

Table 2: Thin film layer selection utilized in DOE of the Si_3N_4 waveguide platform. Root mean square (RMS) roughness and the correlation length of the undercladding (CL), correlation length of the waveguide sidewall along the waveguide direction (CLZ) and correlation length of waveguide sidewall perpendicular to the waveguide direction (CLX) determined by Anton Paar TOSCA 400 Atomic Force Microscope.

	RMS [nm]	CLZ [nm]	CLX [nm]	CL [nm]
Thermal SiO_2 (Undercladding)	0.28	-	-	10.5
HD-PECVD SiO_2 (Undercladding; after CMP)	0.64	-	-	7.1
PECVD sidewall Si_3N_4 (Waveguide)	2.91	126.10	819.96	-
PECVD top-surface Si_3N_4 (Waveguide)	1.98	17.14	29.29	-
LPCVD sidewall Si_3N_4 (Waveguide)	2.04	141.99	1390.52	-
LPCVD top-surface Si_3N_4 (Waveguide)	0.87 nm	104.83 nm	417.37 nm	-

The layer stacks based on the 2x3 full-factorial DOE are summarized for completeness in Table 3.

Table 3: Layer stacks used in the 2x3 full-factorial DOE

	Undercladding	Waveguide layer	Uppercladding
Stack 1	Thermal SiO_2	LPCVD Si_3N_4	HD-PECVD SiO_2
Stack 2	Thermal SiO_2	LPCVD Si_3N_4	Sputtered SiO_2
Stack 3	Thermal SiO_2	PECVD Si_3N_4	HD-PECVD SiO_2
Stack 4	Thermal SiO_2	PECVD Si_3N_4	Sputtered SiO_2
Stack 5	HD-PECVD SiO_2	LPCVD Si_3N_4	HD-PECVD SiO_2
Stack 6	HD-PECVD SiO_2	LPCVD Si_3N_4	Sputtered SiO_2
Stack 7	HD-PECVD SiO_2	PECVD Si_3N_4	HD-PECVD SiO_2
Stack 8	HD-PECVD SiO_2	PECVD Si_3N_4	Sputtered SiO_2

The detailed steps of the fabrication process are illustrated in Figure 3, including the deposition of all the layers and the structuring step.

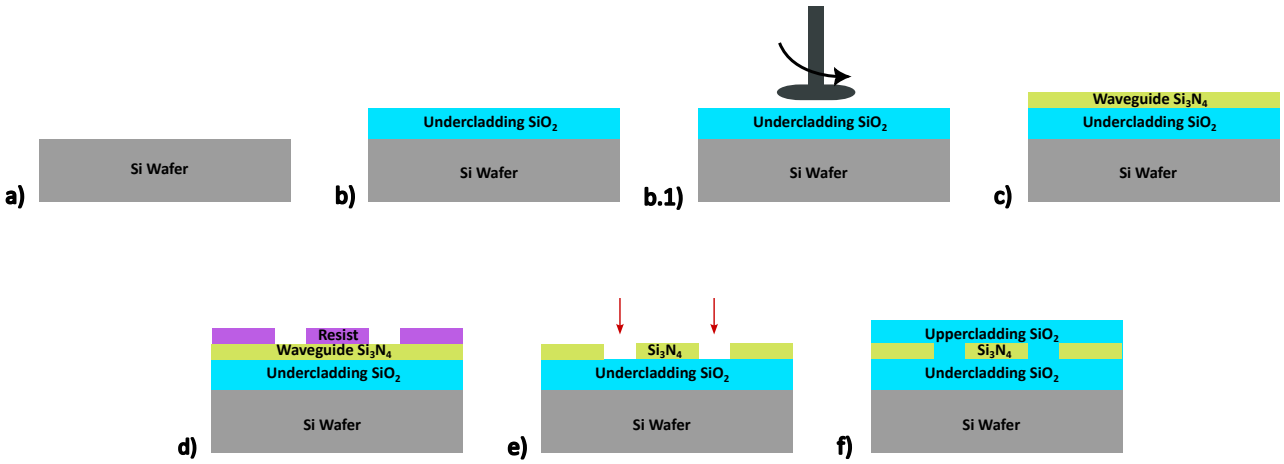


Figure 3: Fabrication flow: a) Bare silicon wafer as substrate; b) Deposition/Growth of SiO_2 as undercladding; b.1) CMP step on undercladding for surface smoothing (only if HD-PECVD SiO_2 is deposited); c) Deposition of Si_3N_4 as waveguide layer; d) Photolithographic resist deposition and exposure; e) Structuring of Si_3N_4 waveguides via reactive ion etching; f) Sputtering or HD-PECVD deposition of SiO_2 as uppercladding.

The DOE utilized a silicon wafer with a thickness of 725 µm ± 15 µm as the substrate material (Figure 3 a)). Before any deposition, the wafer was meticulously cleaned using wet chemistry methods in a weak acidic solution to eliminate any undesired particle contamination. Subsequently, either a 2.4 µm-thick layer of SiO_2 was deposited using the HD-PECVD technique, or a wet chemical oxidation to achieve a 2.2 µm-thick SiO_2 was carried out. This is schematically sketched in Figure 3 b).

To improve the uniformity and minimize the surface roughness for the case of HD-PECVD SiO_2 undercladding, chemical-mechanical polishing was performed, illustrated in Figure 3 c). This polishing utilized a silica-based slurry, lasted 70 seconds and resulted in the removal of a 200 nm layer (Figure 3 b.1). Afterwards, a 250 nm-thick Si_3N_4 waveguide layer was deposited either via LPCVD or PECVD, shown in Figure 3 c).

After each deposition, ellipsometer measurements were employed to monitor thickness and stress. During the structuring phase (Figure 3 d), advanced deep ultraviolet (DUV) lithography was used with state-of-the-art scanner equipment. Following resist critical dimensions (CDs) verification and a manual SEM inspection, a two-step reactive ion etching process (Figure 3 e) was conducted using endpoint signal detection.

After ensuring the removal of resist residuals and confirming the etch CDs through a post-etch SEM analysis, a 2.4 µm-thick uppercladding layer was sputtered, which is beneficial for applications like sensing.[14] Alternatively, HD-PECVD SiO_2 was utilized with a thickness of 2.4 µm (see Figure 3 f).

The final steps involved ellipsometer assessments of the uppercladding's thickness and supplementary wafer stress evaluations, paving the way for the measurements detailed in the subsequent chapter.

4. OPTICAL MEASUREMENT

The testing process for integrated photonic structures at the wafer level demands meticulousness due to the precise alignment requirements of essential components like single-mode fibres (SMFs). This precision ensures effective connection with external sources and detectors, leading to reliable data collection with high statistical validity. Given these requirements, (semi-)automated testing methods using wafer-scale photonic techniques stand out as the best option. To address these complexities, the FormFactor Cascade SUMMIT200 probe system was utilized, which was specifically designed specifically designed for 200 mm wafers and equipped with a stand-alone silicon photonics module. This cutting-edge module ensures the accurate placement of SMFs (sourced from OZ Optics, SMF-633-4/125), ensuring consistent performance throughout the testing phase.

The study used grating grating couplers (GC) with a 10° coupling angle as primary coupling mechanism. An advanced piezoelectric stage was incorporated to enhance coupling efficiency, which allows ultra-fine adjustments in the sub-micron range, ensuring exact alignment to the pre-determined positions of the GCs. The measurement system's optical pathway included a high-end near-infrared (NIR) diode laser with a center wavelength of 852 nm (LP852-SF30), a polarization controller (Thorlabs FPC030) and a versatile power meter suitable for both visible and near-infrared wavelengths (Newport 2101). Several compatible tools were employed throughout the wafer-level testing: the FormFactor Velox 3.0 software, the FormFactor SiPh-Tool and the FormFactor FFI Photonics interface. This cohesive approach ensured smooth hardware operations and efficient data retrieval. A detailed diagram of the experimental arrangement can be found in Figure 4.

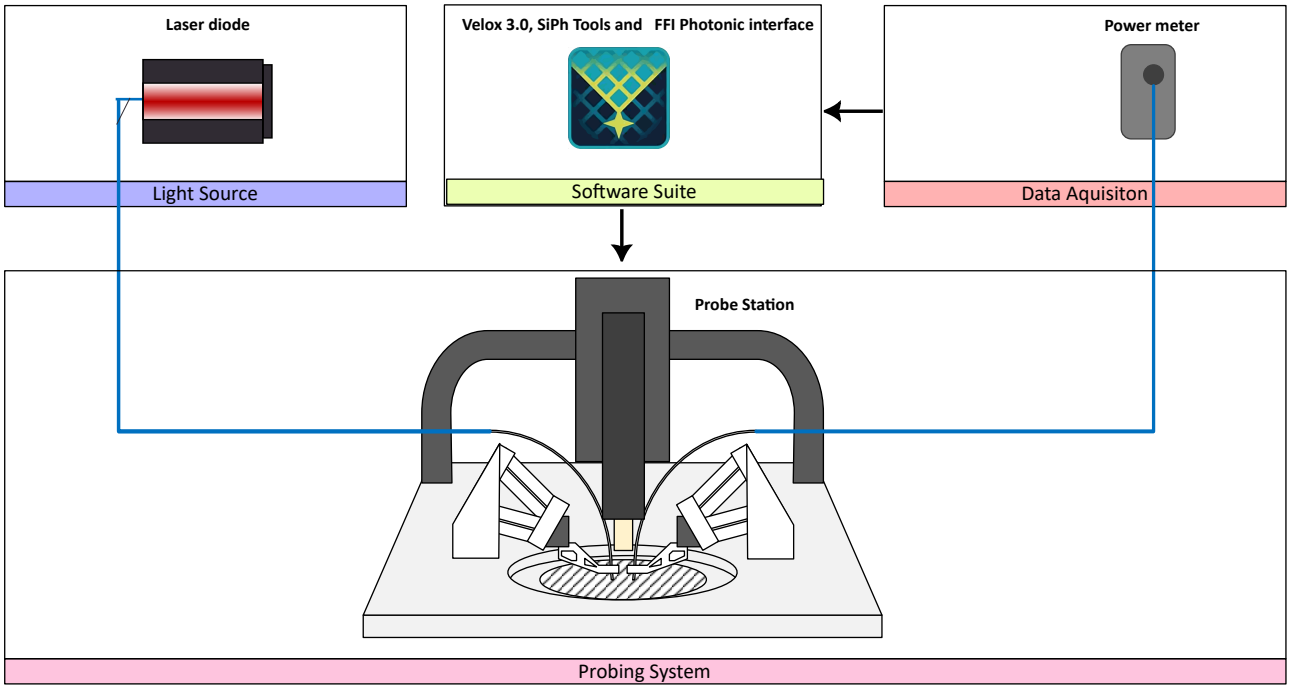


Figure 4: Schematic measurement setup for wafer-level testing of the propagation loss and bend excess loss structures, including probe station, light source, power meter and software control system.

The measurements corresponding to the layout sections shown in Figure 1 and Figure 2 were made on the dies colored in red in Figure 5.

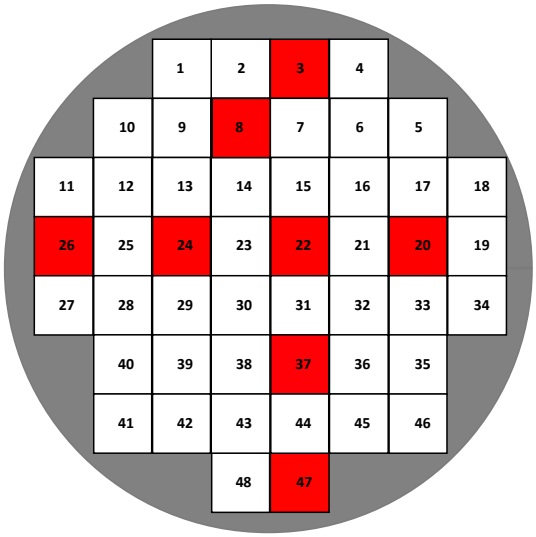


Figure 5: Wafermap of all potential good-dies; Dies marked in red where propagation losses and bend excess losses were determined utilizing the test structures (Figure 1 and Figure 2).

The propagation loss was ascertained using the test structures illustrated in Figure 1, employing the cutback method. The power was gauged across the four waveguide structures with a length disparity of 1.24 cm. The attenuation of this transmitted power was then calculated based on the Lambert-Beer law, represented as:

$$\frac{I(0) - I(l)}{l} = \alpha \tag{1}$$

Here, $I(0)$ and $I(l)$ denote the initial and reduced power in dB, while 'l' signifies the propagation length difference, e.g., in cm. α represents the attenuation constant in dB/cm. This attenuation constant α was derived from a fitting to the aforementioned equation over the test structure length. Any additional losses were factored into the model equation multiplicatively and didn't influence the outcome for the constant α. A pivotal assumption is that all grating couplers have consistent coupling efficiency. This is generally valid in the NIR regime, given known fabrication variations. To enhance the precision of this assessment, all spirals retained an identical number of bends, each with a 50 µm radius, ensuring any discrepancies can be attributed solely to their lengths. This approach also introduces a contingency to counteract measurement inconsistencies. As propagation loss is intrinsically linked to the test structures' length, utilizing spirals of varying lengths facilitates cross-verification, aiding in pinpointing anomalies or ambiguous measurements. The mean values of the propagation losses of all the determined values on the red dies were used for the tree-based modelling.

Bend excess losses share influencing parameters with propagation losses; nonetheless, including bend radius results in mode mismatch-induced losses between straight and bent waveguide sections. Typically, bend excess loss per 90° is evaluated distinctly regarding to the bend radius for every waveguide width. The methodology to gauge bend excess loss is contingent on the designated reference. For the this work, a design was employed with spirals of equal total length but differing bend radii. Every measurement was juxtaposed against a spiral with a bend radius substantial enough to render the bend excess loss inconsequential. Preliminary analyses suggest that a radius of 70 µm can serve as a reference for the waveguides of the test wafer. All other spiral losses were contrasted with this 70 µm spiral, yielding a relative measure known for high reproducibility as long as consistent laser power is maintained throughout the measurement for the group of test spirals. The bend count dictates the cumulative loss per spiral and should be synchronized with the power meter's dynamic range, guided by

empirical or initial measurements. The loss for one 90° bend (BL) was determined for each waveguide width using:

$$BL_x = \frac{I_x - I_{ref}}{n_{bend}} \tag{2}$$

In this equation, I_x represents the measured power in dB for a specific bend radius x. The measured power I_{ref} of the 70 µm reference waveguide in dB is then subtracted and divided by the total number of 90° bends. The grating couplers are assumed to operate with uniform coupling efficiently across all structures. Concerning the bend excess loss structures shown in Figure 2, measurements were acquired from the red dies in Figure 5. For the tree-based modelling, the average propagation loss values derived from the red dies and with their corresponding standard deviations were utilized.

5. EVALUATION OF THE DOE UTILIZING REGRESSION TREE-BASED MODELLING

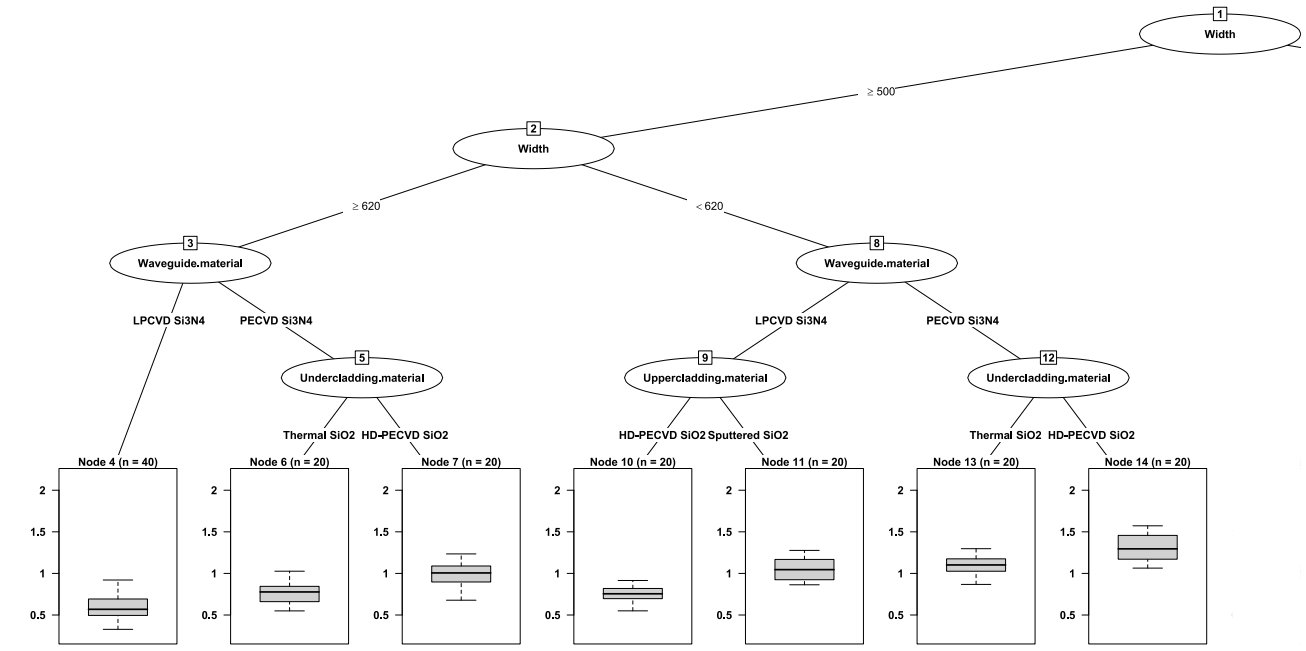
CART was now utilized to evaluate the loss dataset measured from the DOE wafers. Rooted in the simplicity of decision trees, CART can be applied to classification (categorizing data points into classes) and regression (predicting continuous output values), offering a versatile approach to data analysis.[9]

At its foundation, a CART model works by segmenting the dataset into subsets based on the values of input features. This is achieved using a binary tree, where each node represents a decision based on an attribute and the branches represent the outcome of the decision, leading to further nodes or outcomes at the leaf nodes. This binary splitting is a distinctive feature of CART, making it computationally efficient and amenable to several datasets.[15]

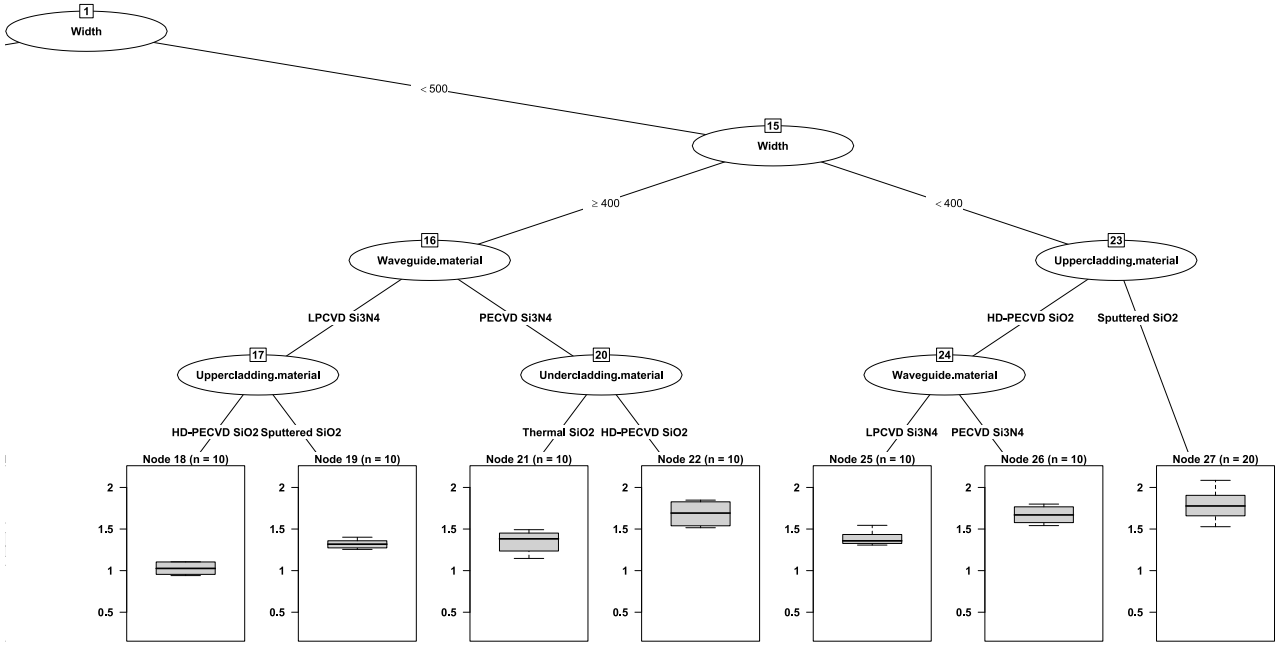
The criteria for splitting the nodes are based on reducing the heterogeneity of data within each subset, using measures like Gini impurity, which is a measure to determine how often a randomly chosen element would be incorrectly classified, for classification tasks and root mean squared error (RMSE) for regression tasks. The RMSE measures the average squared differences between the predicted and actual values. Therefore, it is used to quantify the dispersion or spread of these differences. Lower RMSE values indicate better model performance, with an RMSE of 0 representing a perfect model (i.e., all predictions match the actual values). Both measures guide the algorithm towards making splits that best segregate the data or predict the output, thereby optimizing the accuracy and precision of the model.[9]

5.1 TREE-BASED EVALUATION OF PROPAGATION LOSSES OF THE LAYER-DOE WAFER

In the data mining process, the mean propagation loss on 40 distinct wafers in the DOE, each characterized by six specific width parameters of the waveguides, were evaluated utilizing the statistical evaluation software R (Version 4.3.1) and the rpart-package (Version 4.1.19) for the tree-based evaluation of the 240 observations.[16] In the first step, a regression tree was created using the mean propagation loss over the wafer as the response variable, the layer information and the width as the categorical predictor variable. Pruning the tree to avoid overfitting is a measure to improve the modelling approach. The tree can be seen in Figure 6.



(a)



(b)

Figure 6: Regression tree diagram of the full-factorial DOE layers in Table 3 with the mean propagation loss over the wafer as the response and the layer materials, as well as the width as categorical predictors, created with the rpart-package in R and visualized using the partykit-package tools. Subfigure (a) shows the regression-tree branches for waveguide widths above 500 nm and Subfigure (b) the branches waveguide widths below 500 nm. The terminal nodes represent the mean propagation loss boxplots in dB/cm.

The tree in Figure 6 comprises 27 nodes. The root node (node 1) contains 240 observations, for which it has been identified that the waveguide width is the most significant predictor and suggests a split at 500 nm. The left child node (node 2) root's contains observations where the waveguide width is higher than 500 nm (Figure 6 (b)), while the right child node (node 3) contains observations where the waveguide width is less than or equal to 500 nm (Figure 6 (a)). The second node level again contains the width as the predictor variable where again splitting occurs for waveguides with widths higher than 500 nm into widths higher than 620 nm (750 nm and 650 nm for the teststructure widths) and smaller than 620 nm (590 nm and 550 nm for the teststructure widths) and below 500 nm into the teststructure widths of 450 nm and 350 nm. The third level then splits up regarding the waveguide material for widths larger than 450 nm and 350 nm into the uppercladding material. Here, the branches containing the LPCVD Si_3N_4 waveguides have a lower loss than the PECVD Si_3N_4 waveguides, consistent with the roughness and absorption data in Table 1 and Table 2. In the fourth node level, we have, except for the LPCVD Si_3N_4 structures with a width greater than 620 nm and the nodes for structures with a width of 350 nm, a splitting in the cladding material. For LPCVD Si_3N_4 the uppercladding has a higher influence; for the PECVD Si_3N_4 the undercladding has a higher influence on the tree nodes. According to the roughness data of the undercladding in Table 2 and also possible higher surface defects in HD-PECVD SiO_2 than in thermally grown SiO_2 because the thermally grown SiO_2 is grown from a crystalline SiO_2 and HD-PECVD is chemically grown from precursor gases which are not forming the layer in such an ordered manner. Concerning the higher losses of the sputtered SiO_2 uppercladding compared to the HD-PECVD uppercladding in the node splitting, this is the case that the sputtered SiO_2 deposition step was not optimized, especially for this layout, which might lead to some slight conformality problems, which are lower in HD-PECVD, which is optimized especially for this property.[17,18] The reason for the fact that for LPCVD Si_3N_4, neither cladding is taken as a node for widths of the waveguide greater than 620 nm can be accounted for by the high confinement of the TE-mode at these widths, which, due to the slightly higher refractive index of LPCVD Si_3N_4 even increases the confinement.[13] The reason the uppercladding dominates at 350 nmwaveguide width over the undercladding might be because of the non-conformality in combination with the interaction of the mode on the interface between the waveguide and SiO_2, where there are also air gaps. For PECVD Si_3N_4 at a width of 350 nm, no further splitting is necessary to lower the error. Whereas for LPCVD Si_3N_4, the undercladding also plays a stronger role due to the above mentioned reasons.

In the leaf nodes 4 and 27, the lowest and highest sample medians of the mean propagation loss boxplots were determined in the tree. Node 4 has sample median value of 0.5992 dB/cm, while node 27 has a sample median value of 1.7877 dB/cm. Since the width as a predictor is quite well known to have a tremendous influence on the loss, another regression tree was created, leaving only the material information as categorical predictor variables. This regression tree is depicted in Figure 7.

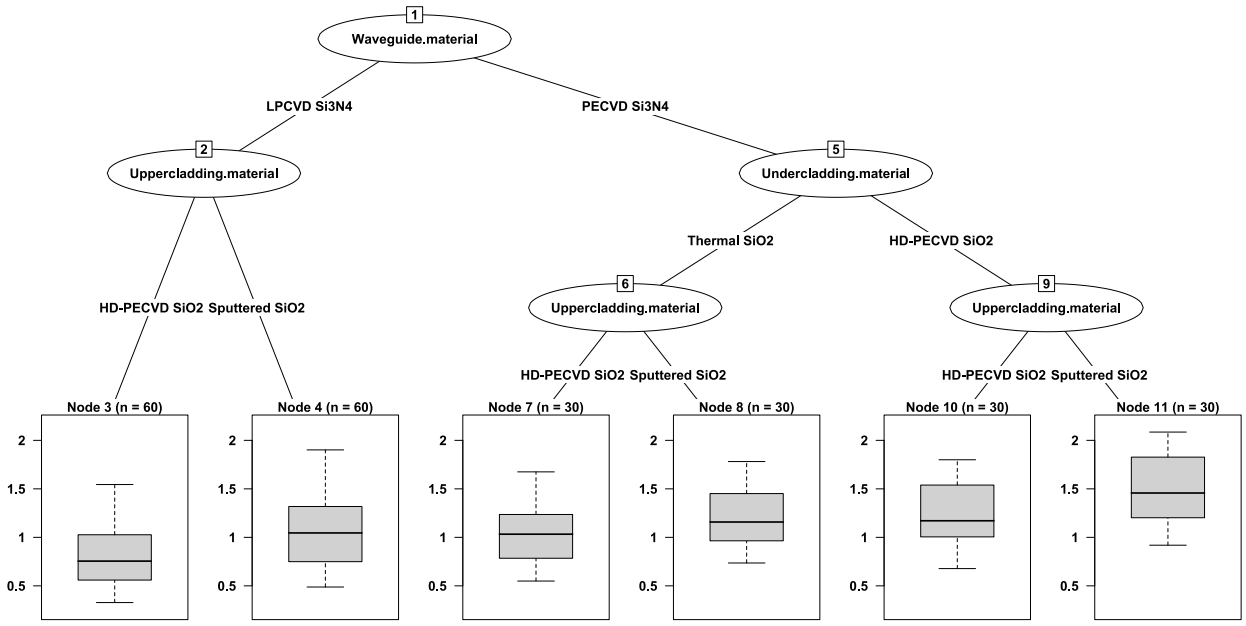


Figure 7: Regression tree diagram of the full-factorial DOE layers in Table 3 with the mean propagation loss over the wafer as the response and the layer materials as categorical predictors, created with the rpart-package in R and visualized using the partykit-package tools. The terminal nodes represent the mean propagation loss boxplots in dB/cm.

In this regression tree, the root node is formed by the waveguide material split up in terms of LPCVD Si_3N_4, where in the next level, the uppercladding material becomes the the pivotal predictor in the second node level and PECVD Si_3N_4 where the uppercladding material becomes the pivotal predictor in the second node level. This can also be observed for medium responses in the leaf nodes in Figure 6. The reason for this, therefore, in both cases might be that for the LPCVD Si_3N_4 waveguides, which have a very low top surface roughness (Table 2), the influence of non-conformality is higher when medium confinement of the mode in the waveguide is the case. For PECVD Si_3N_4 waveguides, which come with a higher surface roughness, the loss is for medium confinement not so much determined by conformality issues of the uppercladding, but more by the undercladding roughness. The splitting at the PECVD Si_3N_4 branch into the uppercladding is again consistent with the observations in Table 1 and Table 2.

In leaf nodes 3 and 11, the lowest and highest means of the propagation loss boxplots were determined in the tree. Node 3 has a sample median value of 0.8219 dB/cm. Node 11 has a sample median value of 1.4782 dB/cm. In the next step, three important measures of both trees in Figure 6 and Figure 7 were compared: the root mean square error of the response, the R^2-value and the RVI of the predictors. The RMSE measures how well the tree's predictions (often the mean response of the observations in a terminal node) match the actual values of the response variable. For each observation in the dataset, the regression tree makes a prediction. This prediction is typically the mean response of all the training data points in the same terminal node as the observation. The prediction error is the difference between this predicted value and the actual value of the observation. In essence, the RMSE provides an average error magnitude between the predicted and actual values.

In regression tree modelling, the apparent R-squared (R^2) statistic is a crucial metric to evaluate the goodness of fit between the model and the data. The apparent R^2 quantifies the proportion of variance in the dependent variable, also known as the target variable, explained by the independent variables integrated into the regression

tree model. It is a valuable indicator of the model's predictive power and ability to capture underlying patterns in the data. Specifically, apparent R^2 is calculated by comparing the total variability in the dependent variable (Total Sum of Squares, TSS) to the variability unaccounted for by the model (Residual Sum of Squares, RSS). A high apparent R^2 value, approaching 1, signifies that the regression tree model effectively accounts for a substantial portion of the variance in the dependent variable, indicating a strong fit. Conversely, a low apparent R^2 value, approaching 0, suggests that the model inadequately explains the variance in the data, implying the need for further model refinement or consideration of alternative approaches. While the apparent R^2 is a valuable measure, it should be complemented with other evaluation metrics and thorough data analysis to comprehensively assess of model performance, particularly when dealing with intricate datasets.

Concerning the variable importance, this measure generally hinges on how often a particular variable is selected to split the data and the improvement in the model's performance due to each of these splits. In the context of a regression tree, performance improvement might be measured concerning a reduction in the RMSE. For instance, a variable that consistently provides a large reduction in RMSE across various tree nodes (or across different forest trees) would be considered more important. Conversely, a variable rarely used for splitting, or that results in minimal RMSE reduction would be deemed less important. Once the importance scores for all variables are obtained, it's often helpful to scale these values to provide relative variable importance scores. The relative variable importance (RVI) can be determined by scaling the importance values to sum up to 1.

The values of the three measures for this work can be seen in Table 4.

Table 4: RMSE, apparent R^2 and RVI of regression tree models in Figure 6 and Figure 7.

Predictor variables	Width + Materials (Figure 6)	Materials (Figure 7)
RMSE [dB/cm]	0.1303	0.3412
Apparent R^2 [1]	0.891	0.269
RVI of waveguide width [1]	0.7485	-
RVI of undercladding material [1]	0.0456	0.1669
RVI of waveguide material [1]	0.1475	0.4976
RVI of uppercladding material [1]	0.0580	0.3353

The table compares two regression tree models based on their RMSE and the relative importance of their predictor variables. The related figure reference of the models is indicated in the table, suggesting visual representations of the models. The first model, illustrated in Figure 6, incorporating both the width and layer materials as predictors yields an RMSE of 0.1303 dB/cm, indicating a relatively accurate fit with respect also to the errors of the mean propagation loss over the wafer, which is in the range 0.1 dB/cm and 0.2 dB/cm. The apparent R^2 also has a value of 0.8912, which is comparably high. Furthermore, within this model, the predictor waveguide width emerges as the predominant factor, having a high relative importance of 0.7485. The other material properties, such as waveguide, uppercladding and undercladding materials, contribute with the importance of 0.1475, 0.0580 and 0.0456, respectively. This again highlights what was noted previously, that at first, the width and the mode confinement-related loss trends play the most significant role. The waveguide material, where the internal absorption and roughness are important material properties, dominates the loss and finally, the cladding material, where the uppercladding has a higher defect-related influence than the undercladding.

Contrastingly, the second model, depicted in Figure 7, exclusively employs various materials information as predictors. This model has a higher RMSE of 0.3412 dB/cm, suggesting a less accurate fit than the first tree-based model, also due to the very low apparent R^2 of 0.2691. Concerning predictor significance for this model, the waveguide material takes precedence with an importance of 0.4976. The uppercladding and undercladding materials follow with a relative importance of 0.3353 and 0.1669, respectively. So, the comparably high RMSE and the low difference of the RVI indicate that the width should somehow be included in the data analysis. Therefore, another analysis step was undertaken, where tree-based models were built in which only observations from distinct widths were considered. Consequently, a model was built only based on 40 observations. Three waveguide widths were investigated, namely 750 nm, 550 nm and 350 nm and regression tree models were built for each width, which can be seen in Figure 8.

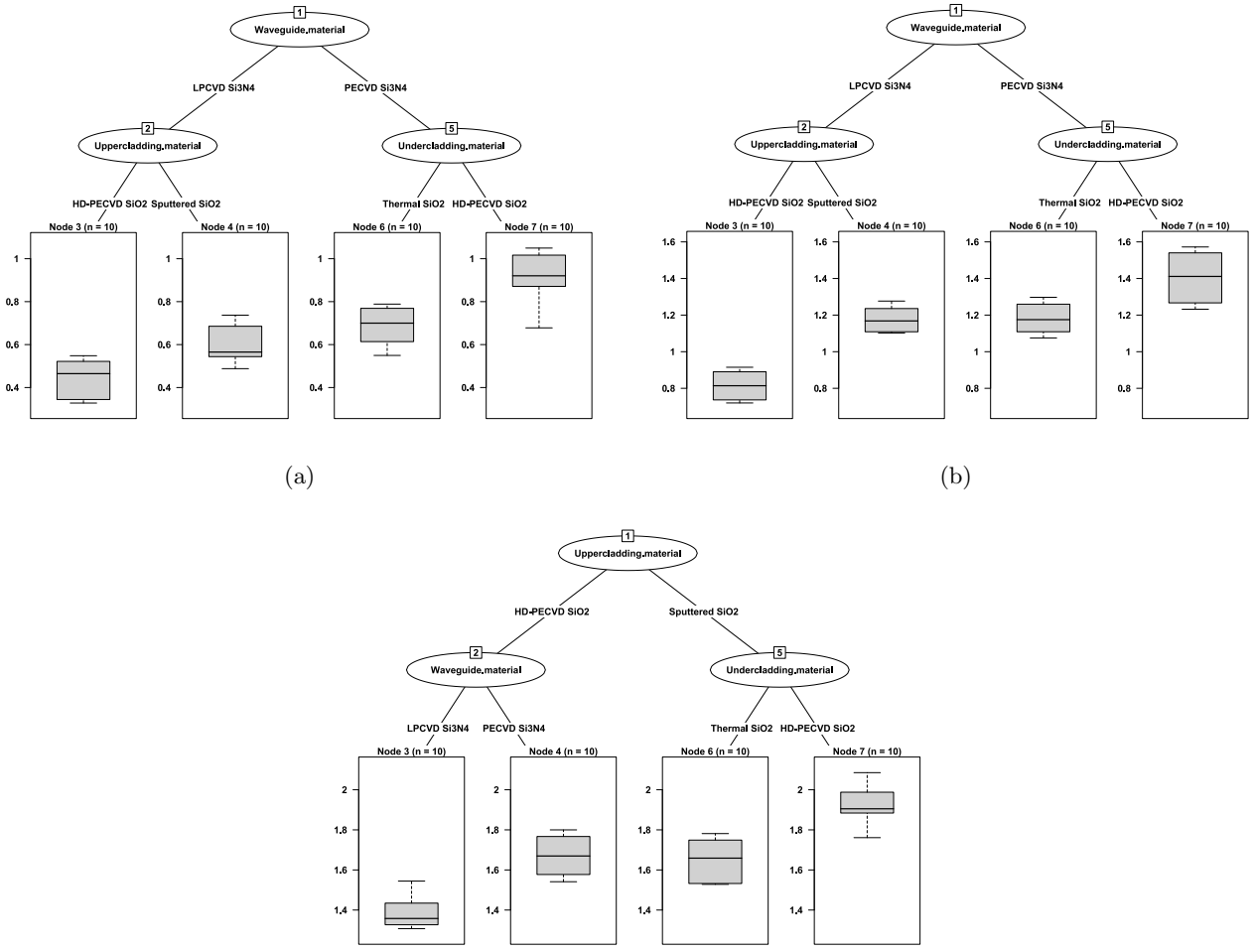


Figure 8: Regression tree diagrams of the full-factorial DOE layers in Table 3 with the mean propagation loss over the wafer as the response and the layer materials as categorical predictors, with the data taken from only a distinct with (40 observations). Subfigure (a) shows the regression tree for waveguides with a width of 750 nm, subfigure (b) for waveguides with a width of 550 nm and subfigure (c) waveguides with a width of 350 nm. The trees were created with the rpart-package in R and visualized using the partykit-package tools. The terminal nodes represent the mean propagation loss boxplots in dB/cm.

When starting the regression tree analysis for a 750 nm width dataset (Figure 8 (a)) with 40 observations, the waveguide material was the dominant predictor. Then the uppercladding is the fundamental predictor for LPCVD Si_3N_4 for the next level and the undercladding for the PECVD Si_3N_4. Among terminal nodes, node 3 has the lowest sample median of the averaged propagation losses (0.4484 dB/cm) and node 7 has the highest (0.9158 dB/cm).

In an analysis of the dataset related to a width of 550 nm, the regression tree looks very similar in terms of the splitting upon the predictors to the tree in Figure 8 (a). However, the terminal nodes reveal further variations. For the 550 nm width dataset, node 3 has the lowest sample median of 0.8135 dB/cm, while node 7 presents the highest sample median of 1.4055 dB/cm. This range is wider than observed in the 750 nm data, suggesting a broader variability in response to the 550 nm width.

From the regression tree model analysis for a 350 nm width dataset, several nuanced insights emerged when considered against the 550 nm and 750 nm widths datasets. The range of mean responses in terminal nodes for the 350 nm data is broader than for the 750 nm but more aligned with the 550 nm dataset. The order of variable importance in the 350 nm dataset is distinct, particularly with the prominence of the uppercladding material, which was also discussed inFigure 6. So the trend with the influence of the cladding for a lower width is also very distinct in these observations For terminal nodes, the lowest sample median response for the 350 nm data is found in node 3, with a value of 1.3860 dB/cm. In contrast, the highest sample median is seen in node 7, pegged at 1.9216 dB/cm. This range of mean responses indicates a relatively wide variability in the dataset for the 3350 nm width. To compare the RMSE, the apparent R^2 and the RVI, the corresponding values of Figure 8 are summarized in Table 5.

Table 5: RMSE, apparent R^2 and RVI of regression tree model

Waveguide width [nm]	750	550	350
RMSE [dB/cm]	0.3412	0.0955	0.0859
Apparent R^2 [1]	0.7859	0.8312	0.8295
RVI of undercladding material [1]	0.2349	0.1374	0.2487
RVI of waveguide material [1]	0.654	0.5009	0.3038
RVI of uppercladding material [1]	0.111	0.3616	0.4473

The table presents an intricate evaluation of the regression tree models over three distinct waveguide widths: 750 nm, 550 nm and 350 nm. From the RMSE values, which illustrate the model's prediction accuracy, it can be observed that the model for the 750 nm waveguides yields the highest error with a value of 0.3412 dB/cm. Conversely, the 550 nm and 350 nm models display enhanced accuracies, documented by their lower RMSEs of 0.0955 dB/cm and 0.0859 dB/cm, respectively. Assessing the apparent R^2 value provides insights into the models' fit, with all models having good fits to the data, especially the 550 nm model, which boasts the highest value of 0.8312, marginally surpassing the 350 nm's 0.8295 and the 750 nm's 0.7859.

The relative importance of the materials reveals nuanced interactions contingent on the waveguide width. The undercladding material's prominence peaks in the 350 nm model with an importance of 0.2487, moderately diminishing in the 750 nm model to 0.2349 and further declining in the 550 nm model to 0.1374. The waveguide material, however, is most influential in the 750 nm width, accounting for a substantial 0.654 of the model's decisions. This dominance reduces to 0.5009 for the 550 nm width and further wanes to 0.3038 for the 350 nm width. The inverse relationship between waveguide material and uppercladding material's importance is strikingly evident. As the waveguide material's importance subsides from 750 nm to 350 nm, the uppercladding material's significance escalates, peaking at 0.4473 for the 350 nm width, compared to the intermediate 0.3616 for the 550 nm and the modest 0.1111 for the 750 nm.

In essence, it has been demonstrated that by reducing the width and thus decreasing the confinement, the waveguide material becomes a less significant predictor than the cladding material, to be more precise, the uppercladding, which dominates as a predictor.

5.2 TREE-BASED EVALUATION OF BEND EXCESS LOSSES OF THE LAYER-DOE WAFER

In the second part, the mean bend excess loss across 40 distinct wafers in the DOE, each defined by four specific waveguide width parameters and eight radii parameters according to Figure 2 was investigated using the same software packages used for the tree-based analysis of the propagation losses. Initially, a regression tree was formulated with the mean bend excess loss across the wafer as the dependent variable and the layer information, along with the waveguide width, as the categorical predictor variables. To mitigate the risk of overfitting, pruning of the tree was implemented, enhancing the model's reliability. This tree is depicted in Figure 9.

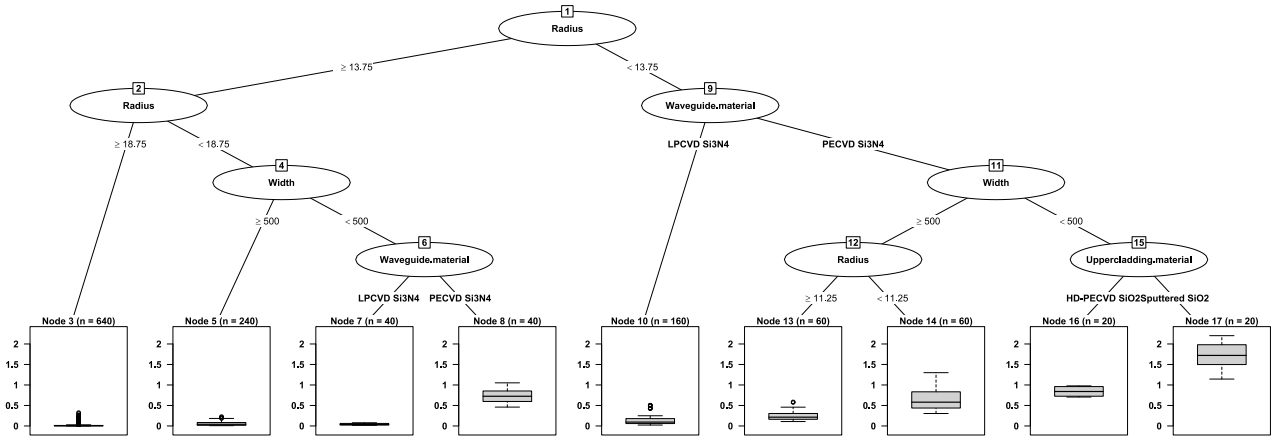


Figure 9: Regression tree diagram of the full-factorial DOE layers in Table 3 with the mean bend excess loss over the wafer as the response and the layer materials, the bend radius, as well as the width as categorical predictors, created with the rpart-package in R and visualized using the partykit-package tools. The terminal nodes represent the mean bend excess loss boxplots compared to the bend spiral with a radius of 70 µm in dB.

The decision tree comprises 15 nodes in total. The root node (node 1), which includes all 1280 observations, indicates that the bend radius of the waveguides is the principal factor affecting some outcomes. The division occurs at 13.75 µm. The root's left child (node 2) captures scenarios less than 13.75 µm, while the right child (node 3) addresses cases with a bend radius of 13.75 µm or more.

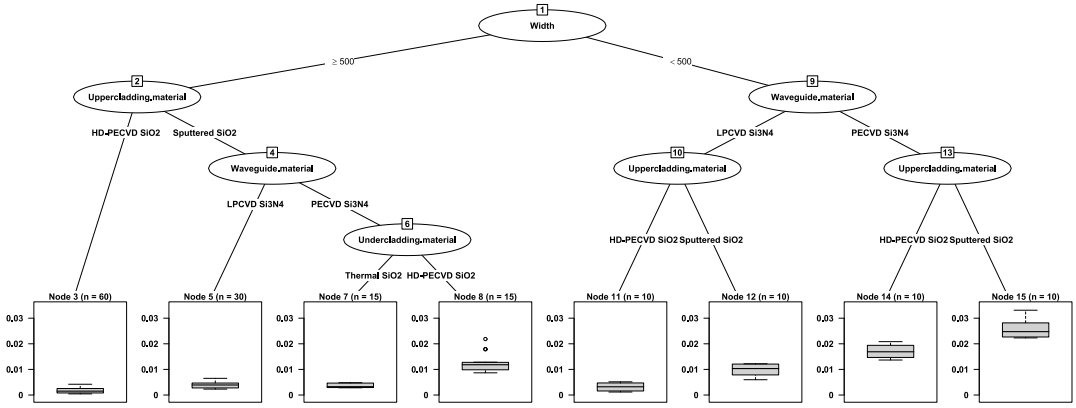
Further down, node 2 bifurcates into two categories: one for a bend radius between 13.75 µm and 18.75 µm and the other for radius of 18.75 µm or above. This node implies a significant distinction in the outcome for different ranges of the waveguide radius, highlighting bend radius as a key determinant.

At the third level, for smaller bend radii (less than 18.75 µm), the subsequent node 3 represents a category without further subdivision, suggesting that, other factors might not significantly affect the outcome within this range of the bend radius. However, for radii between 13.75 µm and 18.75 µm (node 4), the waveguide width becomes the next influential factor, with a dividing threshold again not specified in the input. This indicates that the waveguide width interacts with the bend radius to influence the outcome in intermediate radius sizes.

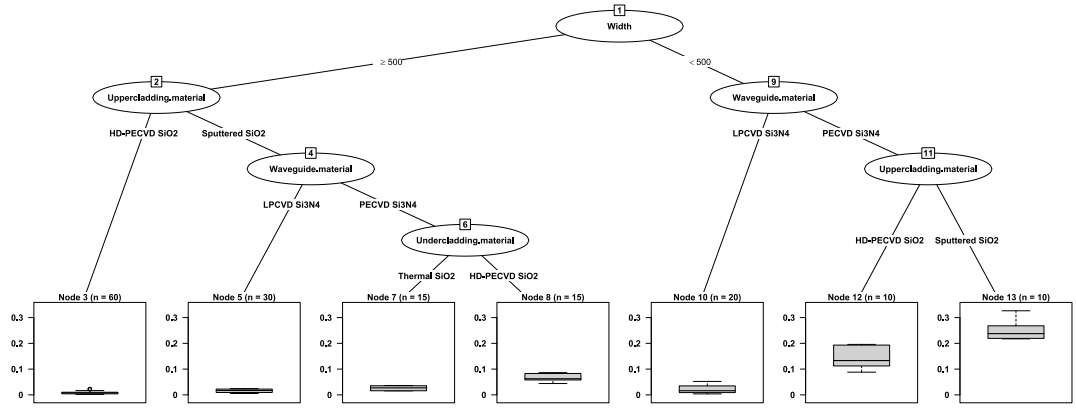
When the radius is larger than 13.75 µm, the waveguide material (node 6), with a split at node 7 based on the waveguide width, demonstrates the interplay between waveguide material and width in larger radii. Interestingly, in this region, the decision tree further divides based on the waveguide width (node 7), pointing to its nuanced impact on the outcome in conjunction with the waveguide material.

In summary, of the 1280 observations in the decision tree, 640 are in leaf node 3, indicating that for radii above 18.75 µm, further splitting is unnecessary due to minimal error reduction and marginal improvement in the apparent R^2 value. An individual tree could be developed for specific radii to detail this node further. Material-wise, PECVD Si_3N_4 exhibits higher losses than LPCVD Si_3N_4, correlating with the roughness and absorption data in Table 1 and Table 2. In node 15, the sputtered SiO_2 uppercladding shows greater losses due to less conformal deposition. Radii below 17.5 µm demonstrate that larger bend radii increase losses. Bend excess losses in Si_3N_4 waveguides are closely tied to mode distribution and confinement, influenced by dimensions like width. Fixed-width waveguides show decreasing bend excess losses with increasing radius. However, larger waveguides lead to greater mode mismatch losses at junctions due to more significant mode dislocation in bends. Thus, both bend radius and waveguide width are crucial in managing mode distribution and confinement, ultimately affecting bend excess losses in Si_3N_4 waveguides.[19–21]

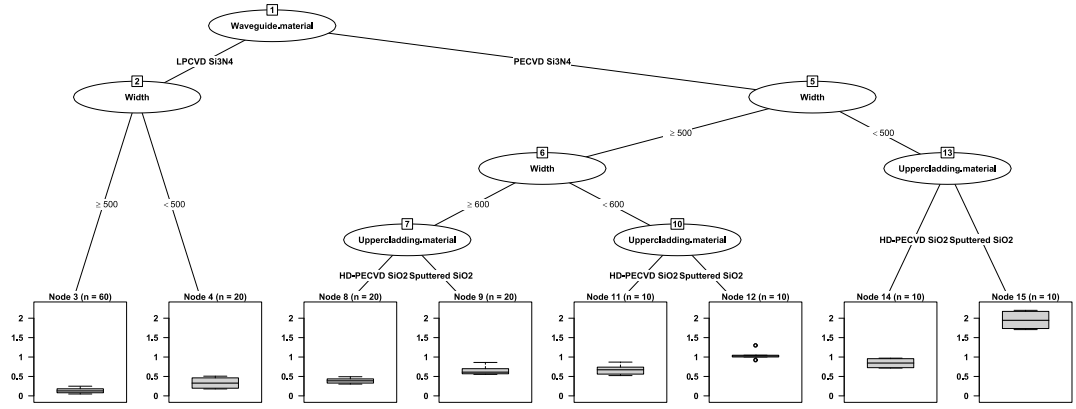
So, in the next step, three bend radius parameters are investigated for the radius of 10 µm, 20 µm and 30 µm. The corresponding regression trees can be seen in Figure 10.



(a)



(b)



(c)

Figure 10: Regression trees of the full-factorial DOE layers in Table 3 with the mean bend excess loss over the wafer as the response and the layer materials as categorical predictors, with the data taken from only a distinct with (160 observations). Subfigure (a) shows the regression tree for waveguides with a bend radius of 30 µm, subfigure (b) for waveguides with a bend radius of 20 µm and subfigure (c) waveguides with a bend radius of 10 µm. The trees were created with the rpart-package in R and visualized using the partykit-package tools. The terminal nodes represent the mean bend excess loss boxplots in dB/cm.

In the given decision tree in Figure 10 (a), where only test structures with a bend radius of 30 µm are involved, the prediction of bend excess loss in Si_3N_4 waveguides is primarily influenced by the geometry, specifically the width of the waveguides and the materials used in the waveguide and claddings. The tree begins with an initial decision point that separates the waveguides based on their width at 500 nm. This suggests that the bend excess loss behaviour changes significantly for waveguides narrower or wider than this threshold.

For waveguides with a width less than 500 nm, the tree indicates that the upper cladding material is the next most influential factor, with the mean bend excess loss reduced compared to the root. This indicates of confinement losses are more significant in this regime; tighter confinement due to the waveguide geometry could lead to increased interaction of the guided mode with the cladding material, hence its importance in the model.

Conversely, for waveguides with a width of 500 nm or greater, the material of the waveguide itself is a major factor. This implies that for wider waveguides, where the mode may be less tightly confined, the intrinsic properties of the waveguide material become more critical in determining the bend excess loss. This can be associated with the transition from high confinement (where the waveguide's material properties dominate the loss mechanisms) to a regime where the mode interacts more significantly with the surrounding materials.

Further splits in the tree refine the bend excess loss prediction based on additional factors like the materials of the upper and under claddings. In particular, for the narrower waveguides, the decision tree's focus on cladding materials after the initial split based on width implies that for these geometries, the interaction of the evanescent field with the cladding materials is a notable contributor to bend excess loss.

Reducing the radius to 20 µm i(Figure 10 (b)) results in a notable change in the relative importances, the model indicates that the material of the waveguide has increased in significance, but still with the width of the waveguide as predominant predictor, reinforcing the fact that the geometry plays a crucial role in determining bend excess loss. However the loss dominating factors of the waveguide layer become more important.

The data at 10 µm bend radius (Figure 10 (c)), the waveguide material emerges as the most significant feature. The uppercladding material gaining importance as the radius decreases, implying a stronger interaction with the target variable at smaller scales. However it does not overtake the width as the second most important factor.

The RMSE, the apparent R^2 and the RVI of the predictors of the regression tree models in Figure 9 and Figure 10 are compared in Table 6.

Table 6: RMSE, apparent R^2 and RVI of regression tree model

Bend radius [μm]	All widths in Figure 2	30	20	10
RMSE [dB]	0.0925	0.0019	0.0166	0.0245
Apparent R^2 [1]	0.9009	0.9237	0.9357	0.9512
RVI of waveguide radius [1]	0.3438	-	-	-
RVI of waveguide width [1]	0.3092	0.4141	0.3693	0.3110
RVI of undercladding material [1]	-	0.082	0.024	-
RVI of waveguide material [1]	0.2709	0.3411	0.4879	0.4731
RVI of uppercladding material [1]	0.0759	0.1619	0.1181	0.2157

This analysis explains how tighter bend radii correlate with increased predictive accuracy and variable sensitivity in the context of bend excess loss, as reflected by the RMSE and apparent R^2 value. The RMSE showed that the model's accuracy improves as the bend radius decreases, with the smallest radius of 10 µm achieving an RMSE of 0 0.0245 dB. This trend is mirrored in the apparent R^2 values, which climbed from 0.9009 for the overarching model to an impressive 0.9512 for the 10 µm radius model, suggesting that models become more aligned with the empirical data at smaller bend radii due to the higher losses involved and the higher relative differences in the loss values.

The findings from this study highlight the complex interplay between waveguide dimensions, material properties and the physical constraints imposed by reduced bend radii. The data indicates that confinement effects, particularly associated with waveguide width and material composition, are pivotal factors in the optimization of bend loss, a critical consideration in designing and developing miniaturized photonic devices.

6. CONCLUSION AND OUTLOOK

A comprehensive evaluation was undertaken into the intricacies of a qualitative analysis of optical loss sources for propagation and bend excess losses in waveguides on Si_3N_4 platform variations at a wavelength of 850 nm.

Wafers with dedicated test structures were fabricated in a 2x3 full-factorial DOE with five wafers per split according to the three layers (cladding and waveguide) and two deposition variations. In addition, six widths were investigated regarding varied propagation losses.

The propagation losses were measured for distinct dies over the wafer and the mean propagation and bend excess losses were determined. Based on these values, several regression tree models were built, with the response being the propagation and bend excess losses. The predictors were varied according to the observations. At first, all possible predictors (width, uppercladding material, waveguide material and undercladding material) were taken into account, revealing the width as the most dominant predictor for the losses, which is because the width heavily influences the confinement and thus also the interaction of the mode in the waveguide with different interfaces. To get more information about the layers' role as predictors, another model was built, not considering the width as a predictor. The fit quality was worse than expected, but it was shown that the waveguide material is, in this case, the most dominant predictor. In the final analysis step, observations made on distinct widths were considered (750 nm, 550 nm and 350 nm). Therefore, it was revealed that the RVI shifts from being the highest for the waveguide material at a width of 750 nm to being the highest for the uppercladding material at a waveguide with of 350 nm, where the interaction of the mode with the cladding is the strongest. The variable importance of the waveguide material continuously decreases due to the decrease in the confinement of the width. For the waveguide layer, the top surface roughness, the sidewall roughness and the absorption usually dominate the loss. For the undercladding the roughness and surface defects play an important role. For the uppercladding the deposition-related defects play a role, especially when sputtered SiO_2 is applied. This is the main influence for the propagation losses for lower waveguide widths and lower confinement.

This second part of the study investigated the factors influencing bend excess loss in Si_3N_4 waveguides using regression tree analysis. The trees revealed that the bend radius and waveguide width are the two most important factors. In the first part, the bend excess loss for all bend radii, widths and materials in the DOE are investigated in one regression tree. The tree revealed that the bend radius is the most important factor in determining bend excess loss, followed by the waveguide width for large radii and the waveguide material for small radii. However especially for large radii a large number of measured data points is in one terminal node. To investigate further, regression trees are built for the constant bend radii 10 µm, 20 µm and 30 µm. At smaller bend radii, the waveguide material becomes more important, while the uppercladding material also plays a role. At larger bend radii, the waveguide width becomes more important, but the waveguide material is a less critical influencing factor, which shows the complex behaviour of the interplay of material-related losses like roughness-induced losses, absoprtion-induced losses and mode-mismatch losses.

The work demonstrates that tree-based modelling can identify vital factors influencing bend excess loss without requiring detailed knowledge of the waveguide structure. This could be particularly useful for optimizing integrated photonic circuits, where it can be difficult to identify the relevant factors affecting the propagation and bend excess loss.

In future work, this approach could be applied in unit process development in Si_3N_4 waveguides fabrication, e.g. improving etching and post-deposition treatments.

Additionally, tree-based models could be combined with other advanced modelling approaches, such as random forests. This could lead to developing new design guidelines for Si_3N_4 waveguides with lower losses and thus, higher figures of merit for biosensing.

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